2026

UK Crime Data
Exploratory Data Analysis

EDA REPORT

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1. Executive Summary

This report provides an exploratory analysis of crime patterns across six UK police force areas to support Nadine Green, Head of Sales, in making location-based property decisions.

The analysis reveals clear differences in crime levels across regions, largely driven by population density and urban structure. Metropolitan areas, particularly London, show the highest crime volumes with strong clustering in central locations, while suburban regions such as Surrey and Thames Valley exhibit lower volumes and more dispersed patterns.

When adjusting for population, the findings highlight that crime risk differs from total crime volume. Regions such as West Midlands and West Yorkshire show higher relative crime exposure, indicating potential variation in perceived safety despite lower overall crime counts than London.

Across all regions, crime is consistently dominated by violence-related and theft-related offences, which are key factors influencing buyer perception, property desirability, and investment appeal. Crime levels remain relatively stable over time, with only moderate seasonal variation.

From a real estate perspective, the analysis demonstrates that crime patterns are highly localised, meaning desirability can vary significantly within regions. Understanding these differences is essential for identifying high-demand areas and avoiding locations with higher perceived risk.

Based on these findings, the Metropolitan Police Service and West Midlands Police are recommended for further investigation, as they provide the most valuable comparison of urban crime patterns and their potential impact on property markets.

2. Introduction

This report presents an exploratory data analysis (EDA) of crime patterns across six UK police forces: City of London Police, Metropolitan Police Service, West Midlands Police, West Yorkshire Police, Thames Valley Police, and Surrey Police. The analysis is designed to support Nadine Green, Head of Sales, in understanding how crime varies across regions and how this may influence property desirability and sales decisions.

The selected police forces represent a diverse range of geographic and demographic contexts. Highly urbanised regions such as London capture areas of high population density and economic activity, while regions such as West Midlands and West Yorkshire represent large metropolitan areas with multiple urban centres. In contrast, Thames Valley and Surrey provide insight into more suburban and semi-rural environments. This variation enables meaningful comparison across different types of property markets.

The dataset consists of police-recorded crime incidents over a two-year period, combined with population data to allow for standardised comparisons through crime rates. Prior to analysis, the data was cleaned and preprocessed to ensure consistency and reliability across all datasets.

The objective of this analysis is to identify key patterns in crime levels, crime types, temporal trends, and spatial distribution, and to assess how these factors may impact location desirability, buyer perception, and property investment potential. The findings will be used to recommend specific police force areas for further investigation in the next stage of the project.

3. Data Dictionary

3.1. Clean Crime Dataset

Column Name	Data Type	Description
crime_id	string	Unique identifier for each crime (may be missing for some records)
month	datetime	Month when the crime was recorded (YYYY-MM format)
reported_by	category	Police force that recorded the crime
falls_within	category	Police force jurisdiction
longitude	float	Geographic longitude of the crime location
latitude	float	Geographic latitude of the crime location
location	string	Description of the crime location (e.g. "On or near High Street")
lsoa_code	category	Lower Layer Super Output Area code (geographic identifier)
lsoa_name	category	Name of the LSOA area
crime_type	category	Type of crime (e.g. burglary, theft, violence)
last_ouctome_category	category	Outcome of the crime (e.g. investigation complete, no suspect identified)
year	integer	Extracted from month
month_num	integer	Numeric month (1–12)

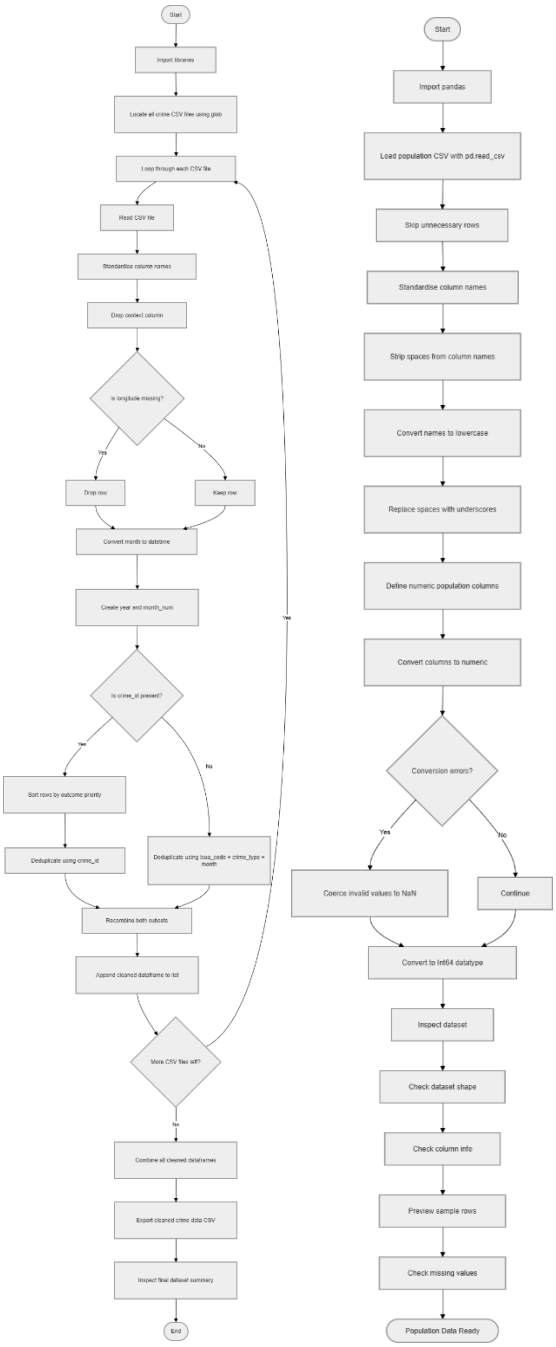
3.2. Clean Population Dataset

Column Name	Data Type	Description
lsoa_code	string	Unique identifier for each Lower Layer Super Output Area (LSOA)
lsoa_name	string	Name of the LSOA area
total	Int64	Total population in the area
f0_to_15	Int64	Female population aged 0–15
f16_to_29	Int64	Female population aged 16–29
f30_to_44	Int64	Female population aged 30–44
f45_to_64	Int64	Female population aged 45–64
f65_and_over	Int64	Female population aged 65+
m0_to_15	Int64	Male population aged 0–15
m16_to_29	Int64	Male population aged 16–29
m30_to_44	Int64	Male population aged 30–44
m45_to_64	Int64	Male population aged 45–64
m65_and_over	Int64	Male population aged 65+

4. Data Cleaning/Preprocessing

Before conducting the exploratory data analysis, a structured preprocessing workflow was developed to ensure that the crime and population datasets were accurate, consistent, and suitable for analysis. Due to the size and complexity of the police crime datasets, preprocessing was first explored on a single police force dataset in a separate preprocessing exploratory notebook. This allowed the cleaning logic, data quality checks, and transformation steps to be tested on a smaller subset before being scaled into a repeatable pipeline for all selected police forces.

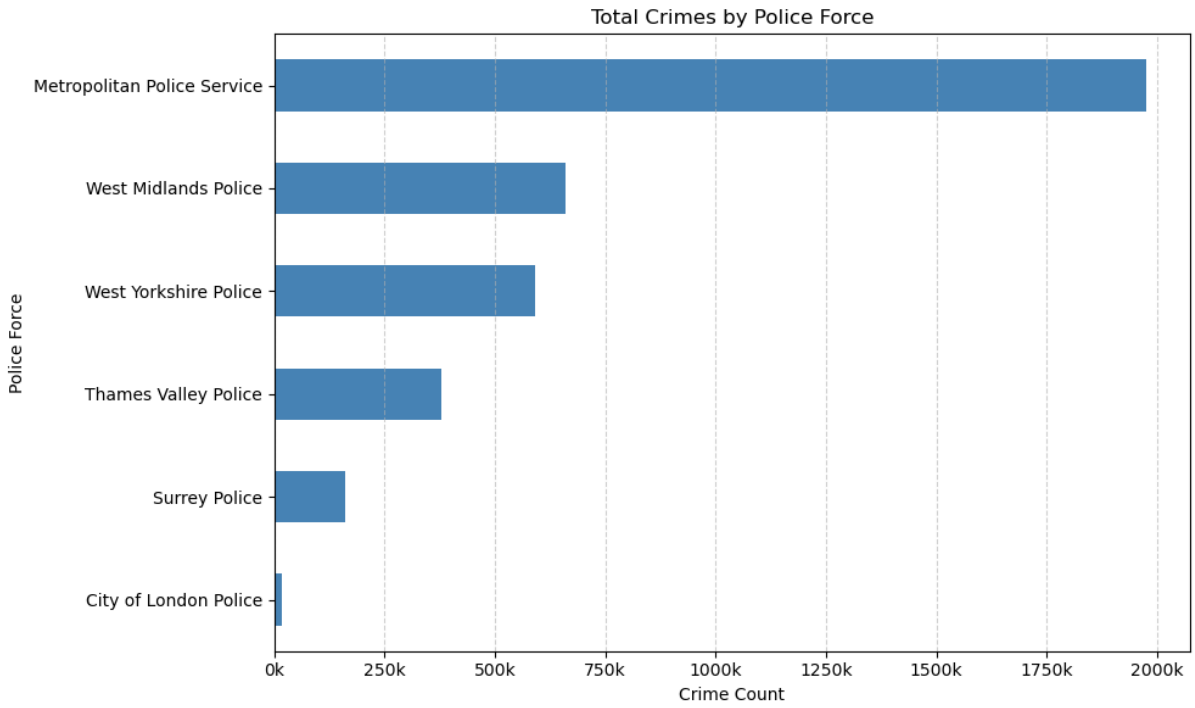
This staged approach improved reliability and reduced the risk of applying incorrect transformations across the full dataset. Once the workflow had been validated, it was implemented in the main preprocessing notebook and applied consistently across all selected crime CSV files.



Preprocessing of Crime and Population Dataset respectively.

5. Exploratory Data Analysis – Number of Crimes

5.1. Number of Crimes in each Police Force Jurisdiction



The Metropolitan Police Service records by far the highest number of crimes, with almost two million incidents, significantly exceeding all other police forces included in the analysis. This is expected as the Metropolitan Police covers Greater London, which has the largest population and one of the most densely populated urban areas in the UK.

West Midland and West Yorkshire Police record the next highest crime totals, with approximately 700,000 and 600,000 incidents respectively. These regions include major metropolitan areas such as Birmingham, Leeds, and Bradford, which tend to experience higher volumes of crime due to larger populations and greater urban activity.

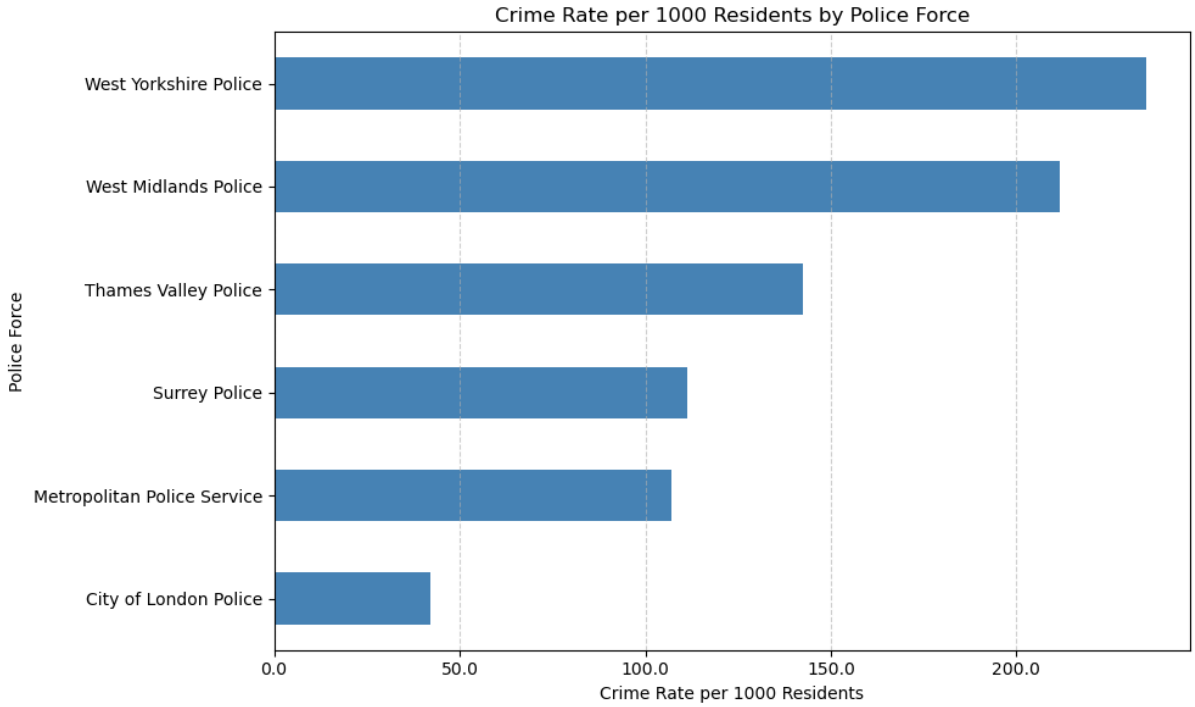
Thames Valley Police record a moderate level of crime incidents, reflecting the mix of urban and suburban environments within its jurisdiction.

Surrey Police records significantly fewer crimes, which may be attributed to its lower population density and more suburban character compared with the larger metropolitan regions.

City of London Police show the lowest total crime count. This is expected because the City of London covers a very small geographic area compared with other police forces.

This chart highlights that total crime counts are heavily influenced by population size and area coverage, meaning they do not necessarily represent the underlying crime risk in each region.

5.2. Crime rate in each Police Force Jurisdiction



This chart standardises crime counts relative to population size. This allows for more meaningful comparison of crime risk across regions.

When adjusting for population, the ranking of police forces changes significantly. West Yorkshire Police have the highest crime rate per 1000 residents, indicating that residents in this region experience a higher relative level of crime compared with other areas.

West Midlands Police also show a relatively high crime rate, reflecting the challenges often associated with large urban environments.

Thames Valley Police and Surrey Police display moderate crime rates, suggesting a more balanced crime environment relative to population.

Interestingly, despite recording the largest number of total crimes, the Metropolitan Police Service has a relatively higher moderate crime rate when adjusted for population. This indicates that the high crime count in London is largely driven by the city's very large population rather than disproportionately high crime risk.

City of London Police again record the lowest crime rate in the dataset. However, this may be influence by the unique demographic characteristics of the City of London, where the daytime population is significantly larger than the residential population.

This comparison highlights the importance of analysing crime rates rather than only absolute crime counts, as it provides a more accurate representation of crime exposure for residents.

5.3. Number of crimes - Key Insights

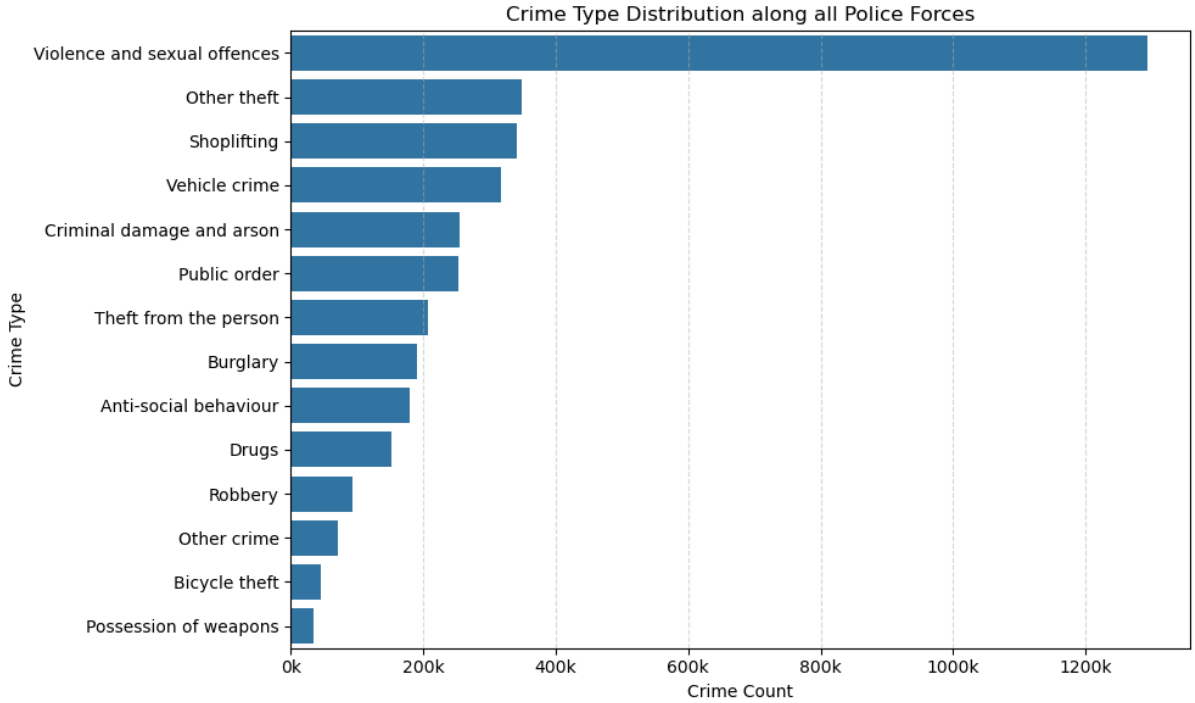
Total crime levels vary significantly across police forces, with metropolitan areas such as London recording the highest volumes due to larger populations and higher levels of economic activity.

However, when adjusted for population, regions such as West Midlands and West Yorkshire show higher relative crime exposure, indicating that crime volume alone does not reflect true risk.

From a real estate perspective, this highlights the importance of focusing on crime rates rather than total counts, as buyer perception, property desirability, and investment decisions are more influenced by relative crime exposure than absolute numbers.

6. Exploratory Data Analysis – Crime Type

6.1. Crime Type Distribution along all Police Forces Jurisdiction



The results indicate that violence and sexual offences are by far the most frequently recorded crimes, significantly exceeding all other categories. This category accounts for a substantially larger number of incidents than any other crime type in the dataset, suggesting that interpersonal offences represent a major component of recorded crime in the analysed regions.

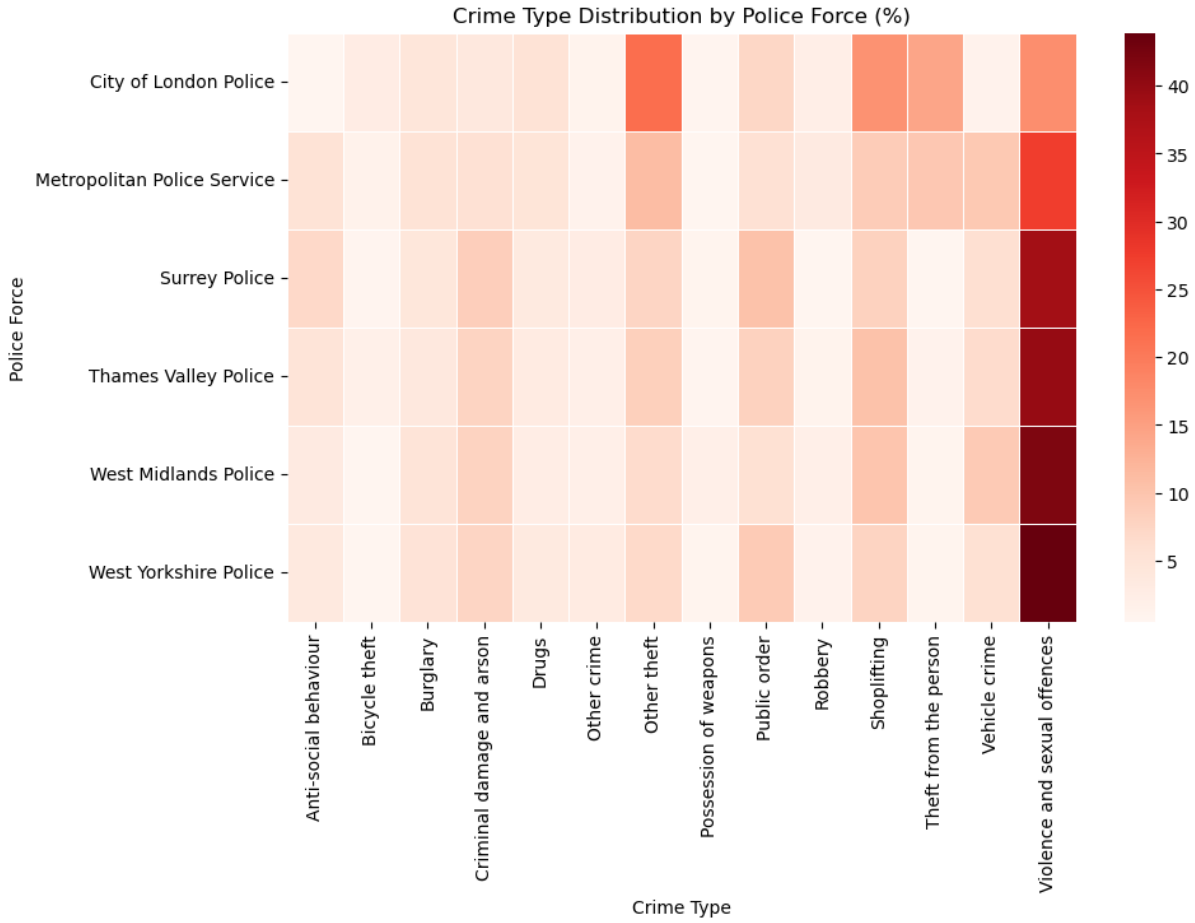
Following this category, other theft and shoplifting represent the next most common crime types. These crimes are typically associated with urban environments and commercial areas, where higher population density and retail activity increase opportunities for theft-related offences.

Vehicle crime and criminal damage and arson also appear frequently in the dataset, indicating that property-related crimes remain an important component of the overall crime landscape.

In contrast, crimes such as bicycle theft, robbery and possession of weapons occur far less frequently. These categories represent a relatively small proportion of the total recorded crimes across all police forces.

Overall, the distribution, suggests that crime incidents are heavily concentrated in a small number of dominant categories, particularly violence-related offences and theft-related crimes.

6.2. Crime Type Distribution by Police Force Jurisdiction



The chart presents a heatmap showing the percentage distribution of crime types within each police force, allowing comparison of crime composition across regions.

Across all police forces, violence and sexual offences consistently represent the largest proportion of recorded crimes, indicating that this crime category is dominant regardless of regional differences. However, the relative contribution of other crime types varies between police forces.

For example, shoplifting and theft-related offences appear more prominent in police forces covering larger metropolitan areas, such as the Metropolitan Police and West Midlands police. This may reflect the presence of large retail districts and higher population density within these regions.

Meanwhile, police forces covering more suburban and semi-urban areas, such as Surrey Police and Thames Valley Police, show a slightly more balanced distribution of crime categories, with theft, public order, and criminal damage contributing relatively similar proportions.

The heatmap also highlights that certain crime types maintain relatively consistent proportions across regions, suggesting common underlying crime patterns throughout the analysed police forces.

6.3. Crime Type - Key Insights

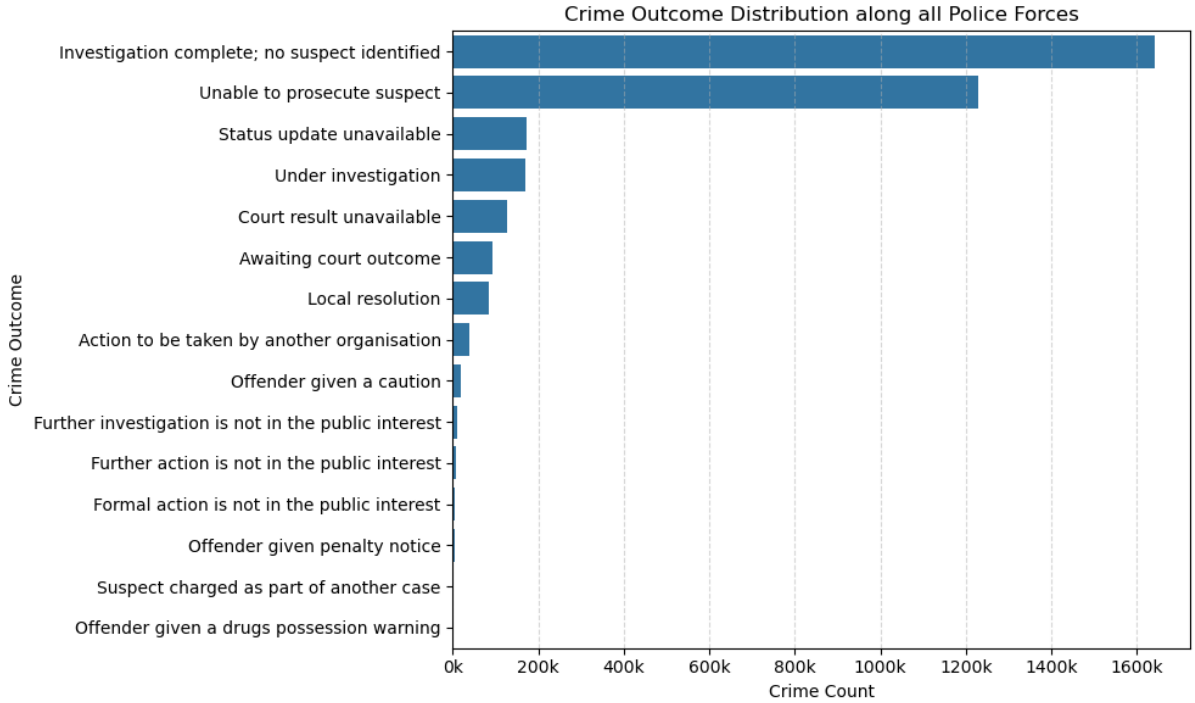
Across all police forces, crime is dominated by violence-related and theft-related offences, with a broadly consistent distribution across regions despite differences in total crime levels.

From a real estate perspective, these crime types are particularly important as they are more likely to influence buyer perception, property desirability, and long-term investment appeal. Areas with higher proportions of these offences may be viewed as less attractive, even if overall crime levels are similar.

Overall, the consistency in crime type distribution suggests that location and crime intensity matter more than crime mix when assessing property attractiveness and sales potential.

7. Exploratory Data Analysis – Crime Outcome

7.1. Crime Outcome along all Police Force Jurisdictions



The data shows the distribution of crime outcomes aggregated across all analysed police forces. The results indicate that the most common outcome is "investigation complete: no suspect identified", which accounts for the largest proportion of recorded outcomes. This suggests that a significant number of cases are closed without identifying a suspect, which may reflect challenges in obtaining sufficient evidence or identifying offenders.

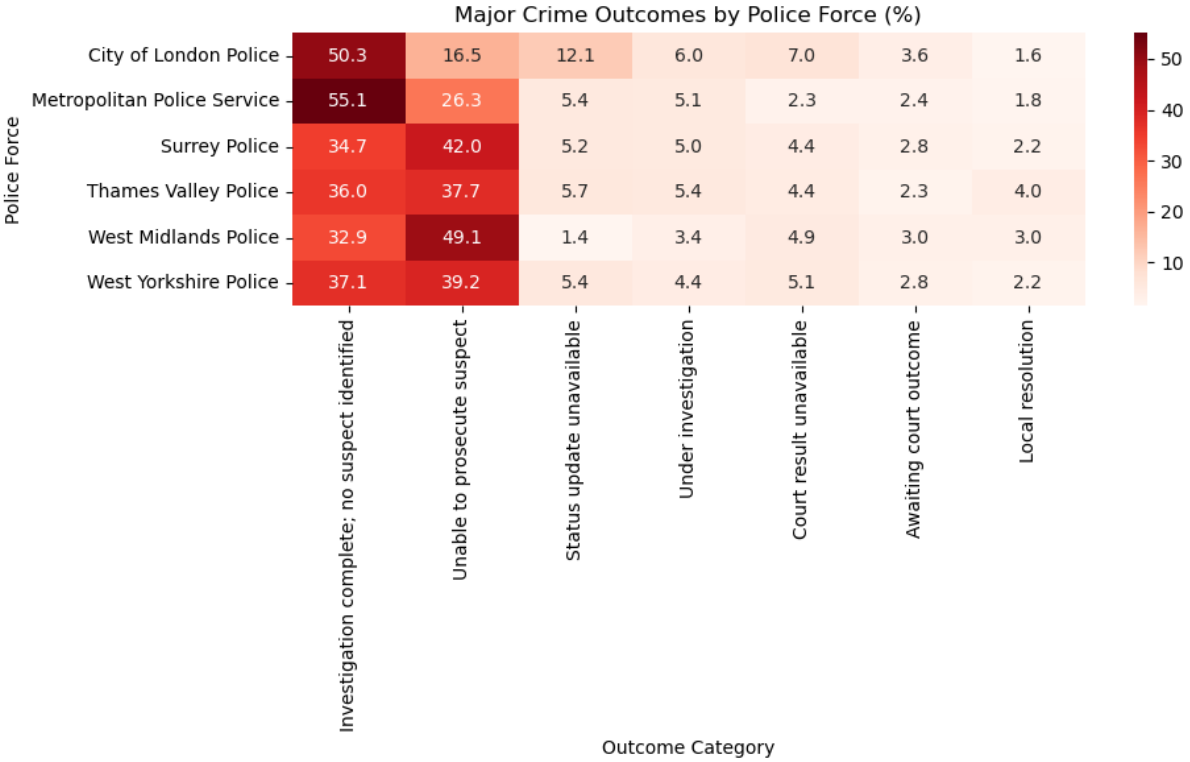
The second most common outcome is "Unable to prosecute suspect/" This outcome indicates that although a suspect may have been identified, there was insufficient evidence to pursue a legal prosecution.

Several other outcomes appear at much lower frequencies, including "Under investigation", "Court result unavailable", and "Awaiting court outcome". These represent cases that are still progressing through the criminal justice system.

Outcomes such as "offender given a caution", "penalty notice", or 'local resolution" represent a relatively small share of cases, indicating that only a limited proportion of recorded crimes result in formal enforcement actions.

Overall, the distribution highlights that a large proportion of reported crimes do not result in a confirmed suspect of formal legal outcome, which provides important context when interpreting crime statistics.

7.2. Crime Outcome percentage in each Police Force Jurisdictions



The data presents a heatmap showing the percentage distribution of major crime outcomes across police forces, allowing comparison of how cases are resolved in different regions. Across most police forces, “Investigation complete; no suspect identified” remains the most common outcome, particularly for the Metropolitan Service and the City of London Police. This suggests that identifying suspects may be more difficult in highly urban environments where crime volumes are larger, and incidents may involve unknown offenders.

However, there are notable regional differences in outcome distribution. For example, West Midlands Police show a particularly high proportion of cases classified as “Unable to prosecute suspect,” indicating that although suspects may be identified, prosecution is often not pursued due to insufficient evidence.

Police forces such as Surrey Police and Thames Valley Police show slightly more balanced outcome distributions, with somewhat lower proportions of cases ending without a suspect identified and a slightly higher share of cases progressing further in the justice process.

The heatmap also highlights that local resolution outcomes remain relatively small across all regions, suggesting that informal resolutions represent only a minor share of crime outcomes in the analysed dataset.

7.3. Crime Outcome - Key Insights

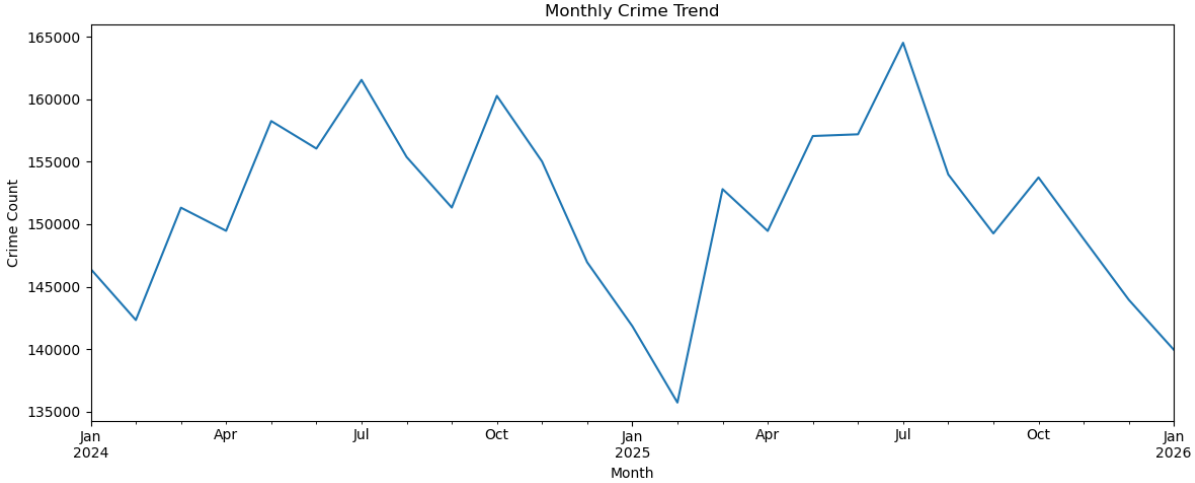
Across all police forces, the majority of crimes result in outcomes where no suspect is identified, or cases cannot proceed to prosecution. This pattern is consistent across regions, with only a small proportion of incidents leading to formal enforcement actions.

From a real estate perspective, this may influence buyer perception of safety and confidence in local policing, particularly in areas with higher crime volumes. Even if crime levels are comparable, differences in outcomes can affect how secure a location is perceived to be.

Overall, crime outcomes provide important context beyond crime volume, highlighting that perceived safety and resolution effectiveness can play a key role in property desirability and buyer decision-making.

8. Exploratory Data Analysis – Time Analysis

8.1. Monthly Crime Trend Across Two Years



This graph shows the total number of recorded crimes, aggregated monthly across all analysed police forces.

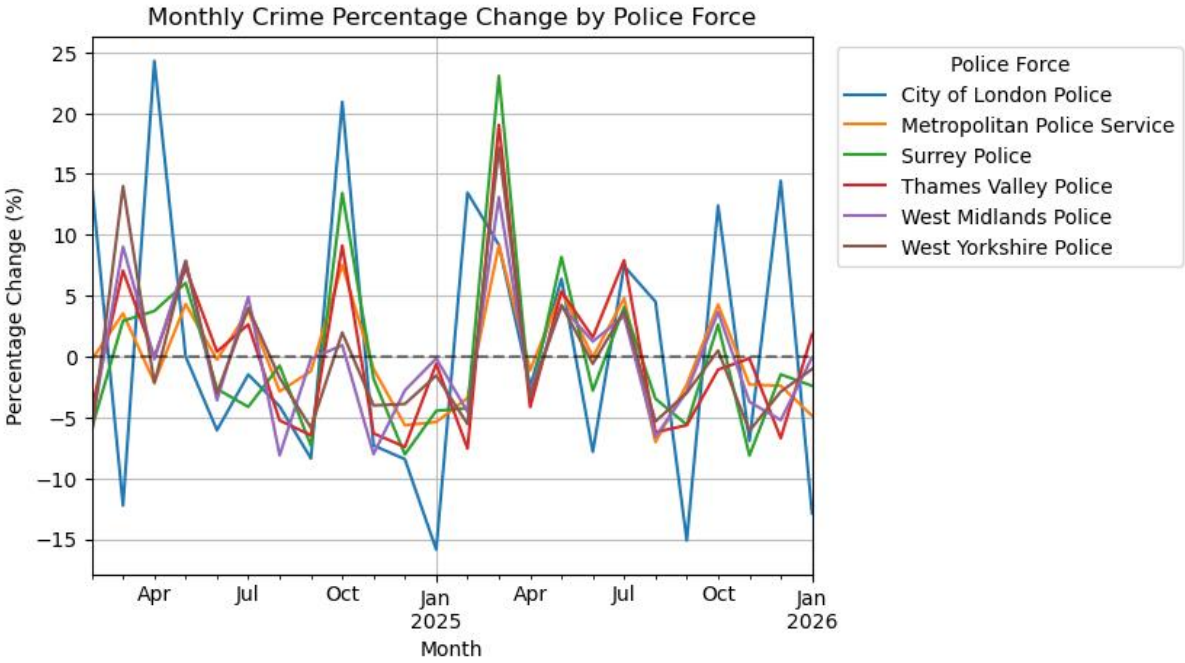
The chart shows that crime levels fluctuate moderately over time but generally remain within a relatively stable range. Monthly crime counts typically range from approximately 140,000 to 165,000 incidents, indicating that while crime levels fluctuate from month to month, there are no extreme long-term increases or decreases over the analysed period.

A noticeable dip in crime levels occurs during early 2025, when recorded incidents drop to the lowest point in the dataset. This may reflect seasonal variation, reporting differences, or short-term fluctuations in crime activity.

Following this decline, crime levels increase again during mid-2205, reaching a peak around June-July 2025, which represents the highest point in the time series. This suggests that crime activity may increase during summer months, potentially due to higher levels of social activity, tourism, and outdoor movement.

Overall, the time series suggests moderate seasonal variation, with crime levels tending to increase during warmer months and decline slightly during winter periods.

8.2. Monthly Crime Percentage Change in Each Police Jurisdiction



The figure above shows the month-to-month percentage change in crime levels for each police force. This allows comparison of how crime trends fluctuate across different regions. Across all police forces, the percentage changes generally fluctuate within a relatively moderate range of approximately ± 10%, suggesting that crime levels tend to change gradually rather than experiencing extreme monthly volatility.

Several police forces experience occasional larger increases or decreases, mainly around the beginning of 2025, where some regions show sharper declines followed by recovery in subsequent months. These variations may reflect temporary changes in reporting patterns, seasonal behaviour, or specific regional events.

Despite these fluctuations, the overall pattern across the police force remains relatively consistent, with most regions showing similar directional movements in crime levels over time. This suggests that broader social or seasonal factors may influence crime trends across multiple regions simultaneously.

8.3. Time Analysis - Key Insights

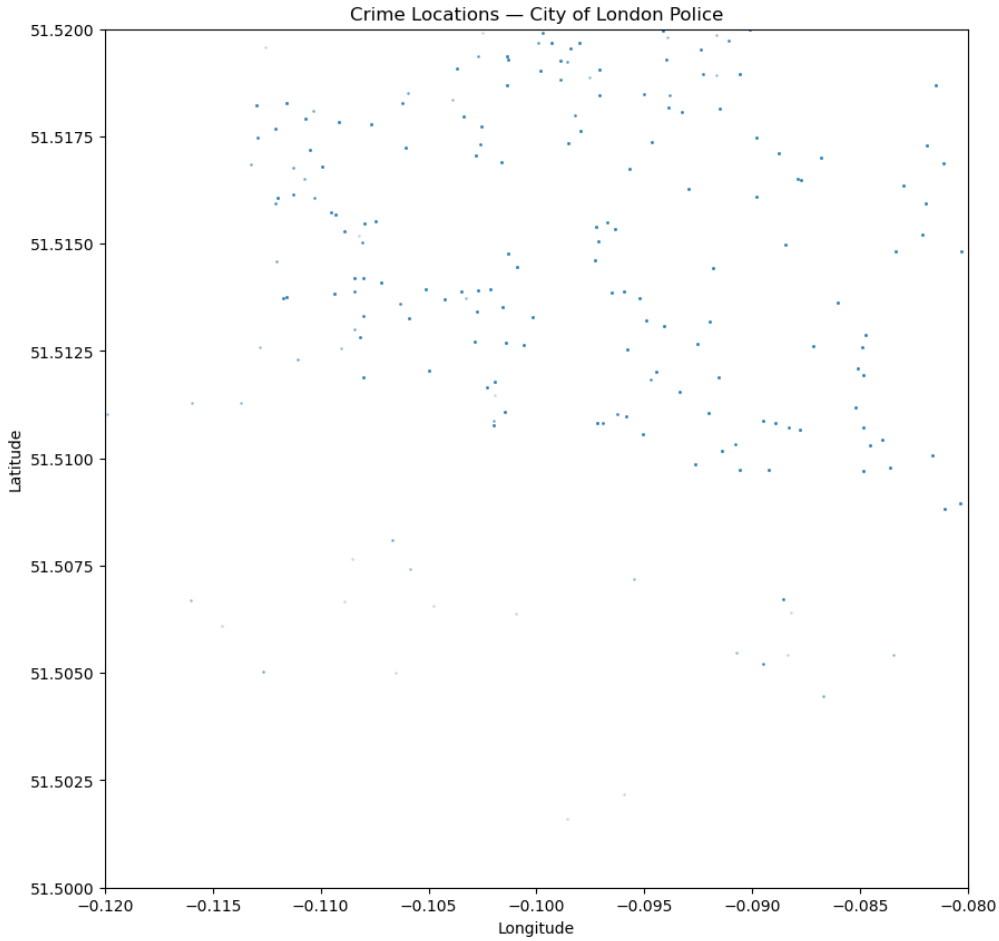
Crime levels remain relatively stable over time, with moderate seasonal variation and slight increases during mid-year months. This suggests that crime patterns are predictable rather than volatile across the analysed regions.

From a real estate perspective, this stability indicates that crime is a consistent factor influencing property desirability, rather than a short-term fluctuation. Seasonal peaks, particularly during summer, may temporarily affect buyer perception in certain areas, but do not indicate long-term changes in risk.

Overall, time-based trends reinforce that location and crime type are more influential than timing when assessing property attractiveness and investment potential.

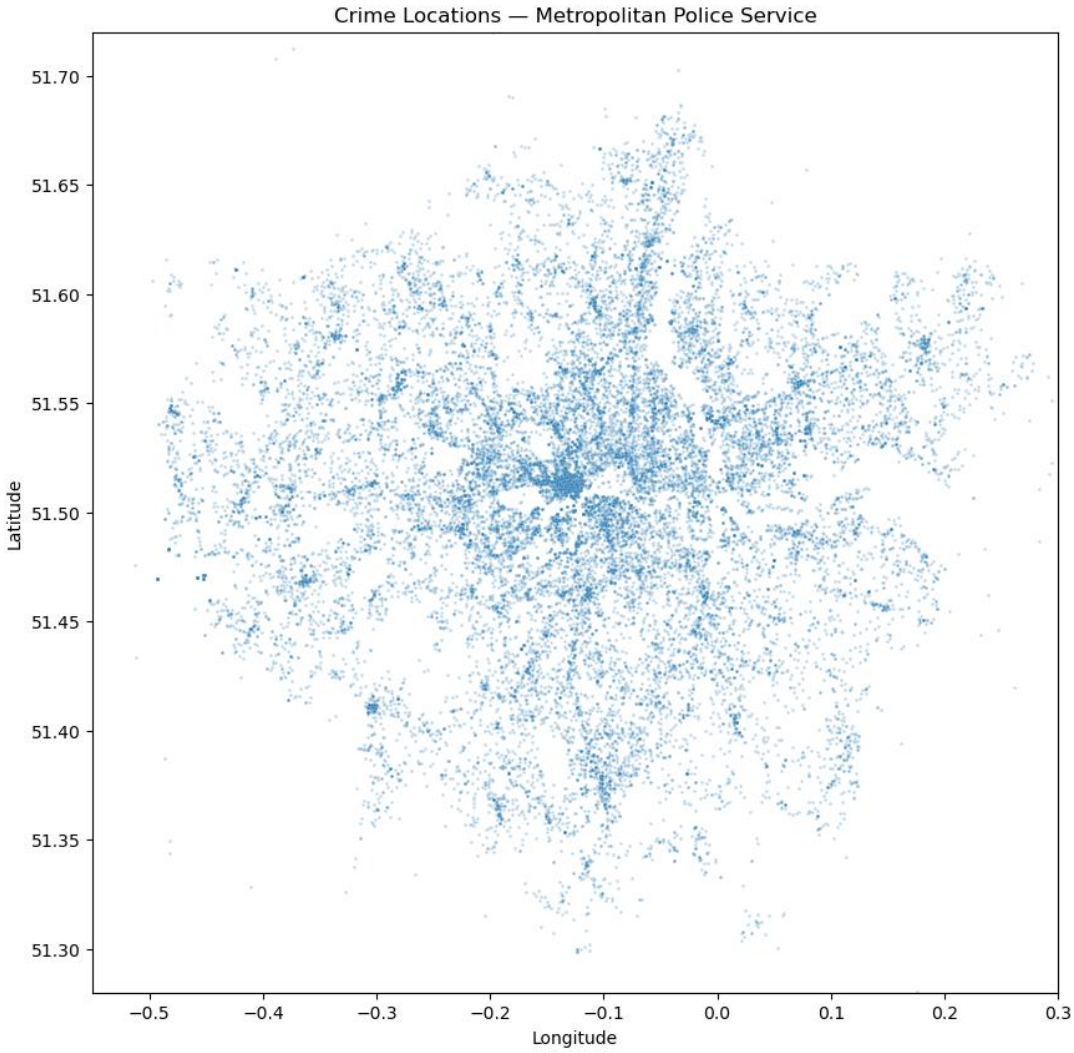
9. Exploratory Data Analysis – Crime Map

9.1. City of London



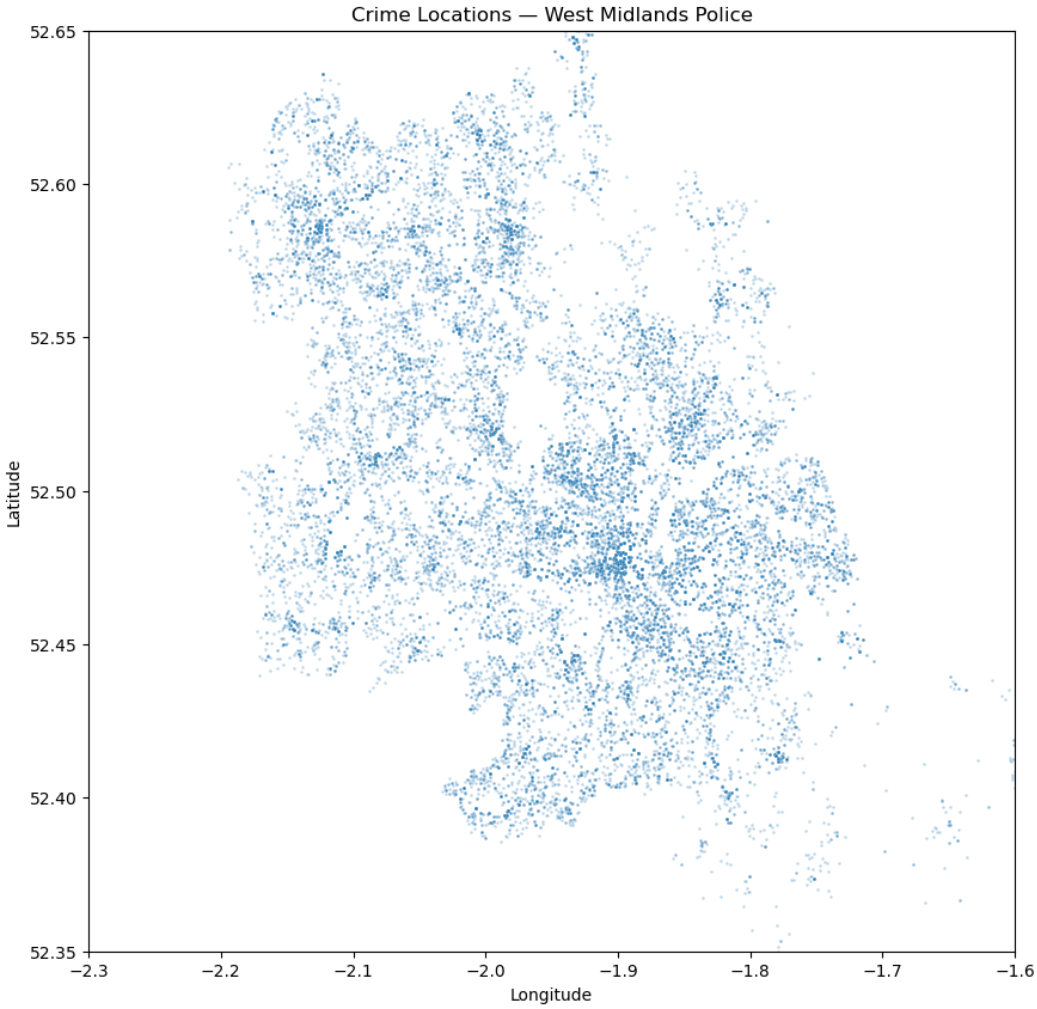
The City of London map shows that there are very few concentrated crime incidents clustered, which supports the crime volume of the region. Crime locations appear concentrated around the central north of the River Thames, where high levels of commercial activity and tourism may contribute to increasing opportunities for certain crime types such as theft and shoplifting.

9.2. Metropolitan Police Service



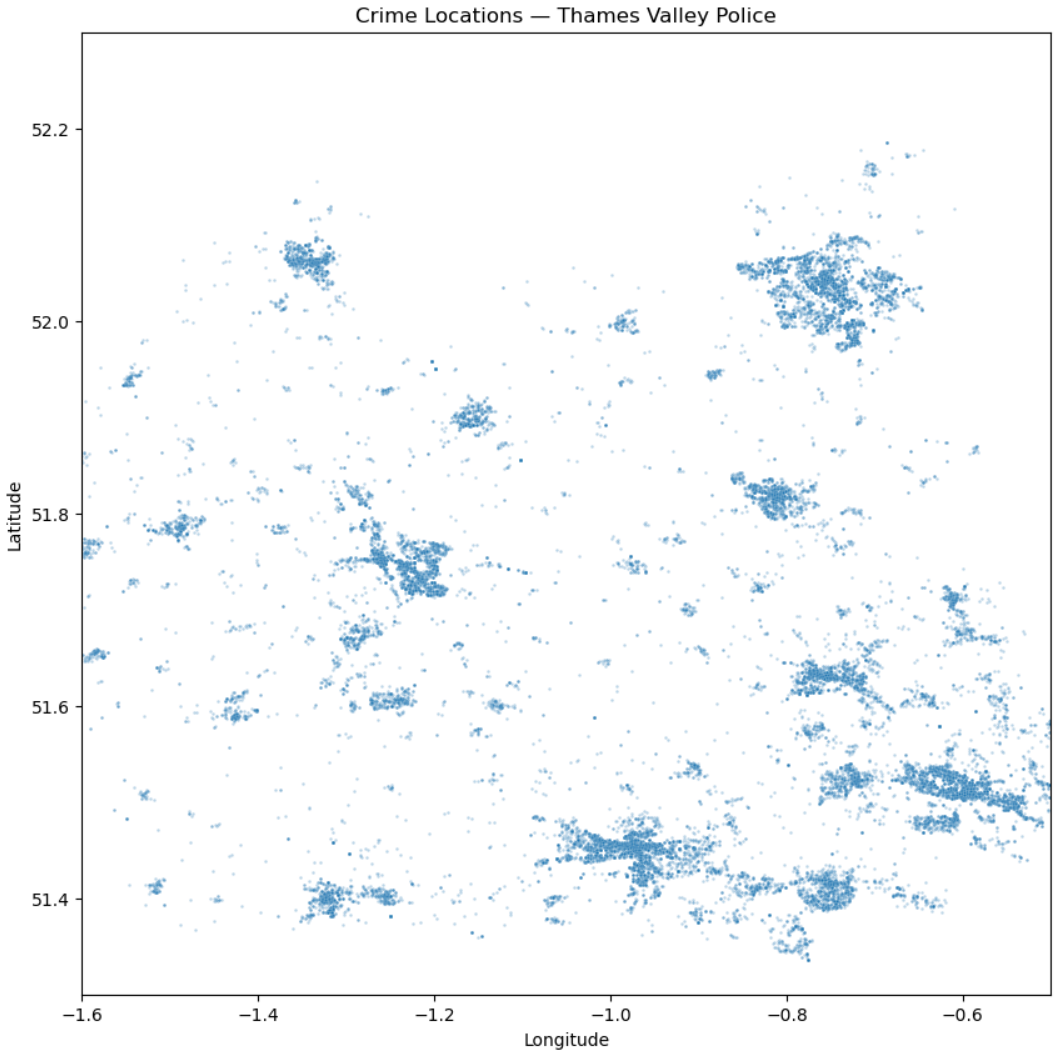
The Metropolitan Police map shows a dense and extensive distribution of crime incidents across Greater London. Crimes are heavily concentrated in central areas, with clusters extending outward along major urban corridors. The high density of points reflects the large population and high level of economic and social activity within the capital.

9.3. West Midlands Police



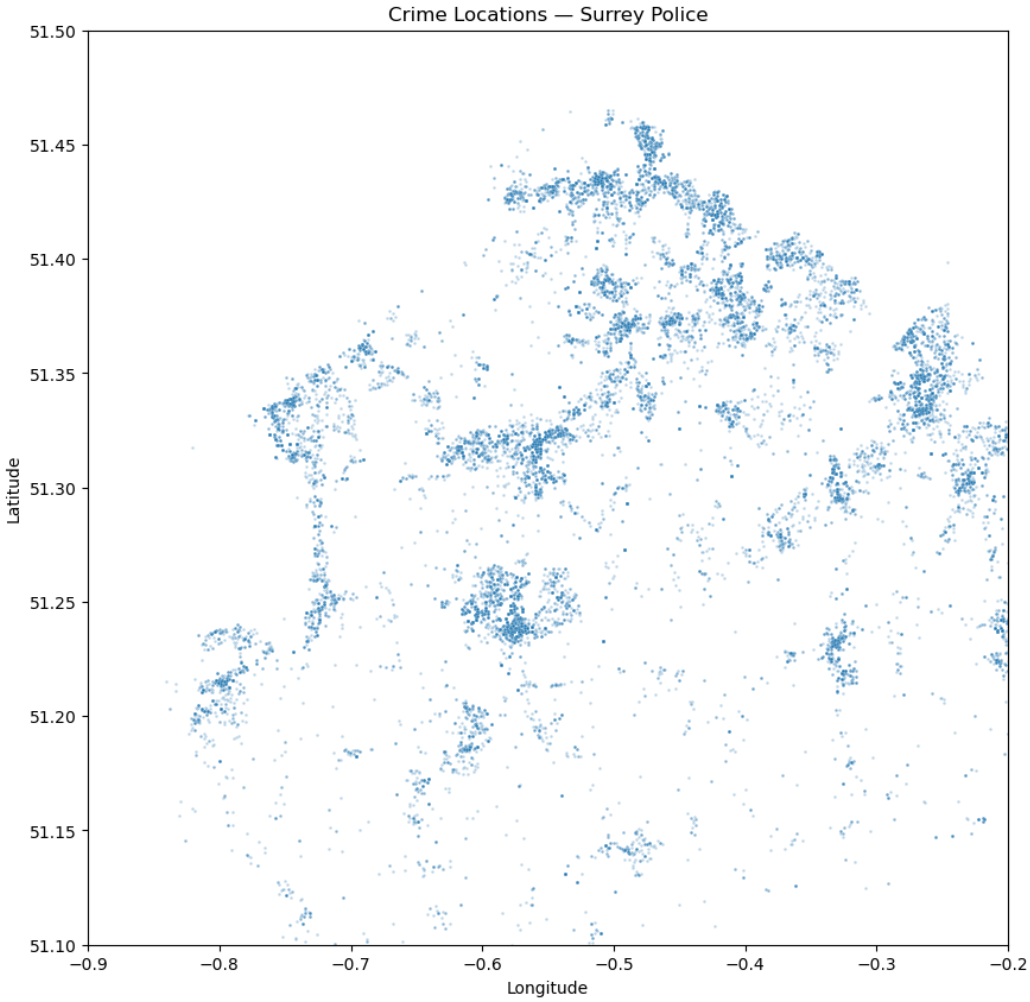
The spatial distribution of crime in the West Midlands region shows several clusters concentrated around major urban centres, particularly the Birmingham metropolitan area. Crime incidents are distributed across multiple neighbouring districts, indicating that crime activity is spread across several densely populated urban areas rather than a single focal point.

9.4. Thames Valley Police



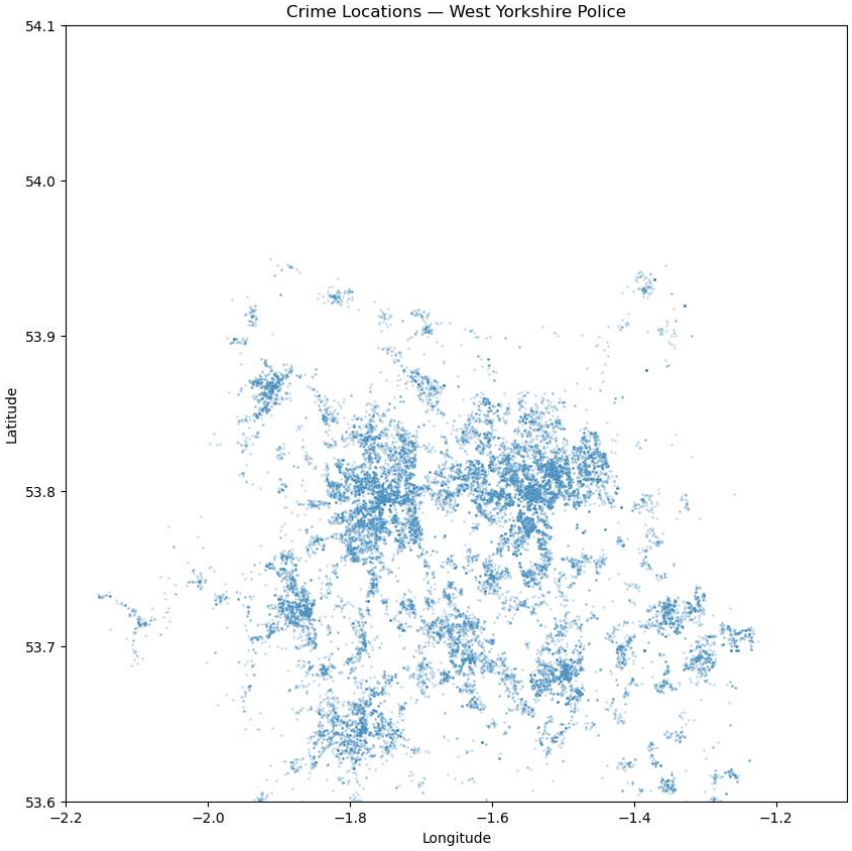
The Thames Valley Police map shows multiple distinct clusters of crime across the region. These clusters likely correspond to major towns and urban centres within the Thames Valley area. Compared with highly urban regions such as London, crime incidents appear more dispersed, reflecting a mixture of suburban and semi-rural environments within this police force area.

9.5. Surrey Valley Police



Crime locations within the Surrey Police area appear more spatially dispersed, just like other suburban areas, with several smaller clusters spread throughout the region. This pattern is consistent with Surrey's predominantly suburban and semi-rural landscape, where crime incidents tend to occur around town centres rather than forming a single large urban concentration.

9.6. West Yorkshire Police



The West Yorkshire Police map shows several distinct clusters of crime concentrated around major urban centres, particularly in areas corresponding to Leeds, Bradford, and surrounding towns. Crime incidents appear densely concentrated in these urban areas, while more rural parts of the region show lower incident density. This spatial distribution reflects the urban structure of West Yorkshire, where several large towns contribute to regional crime patterns rather than a single dominant city centre.

9.7. Key Spatial Insight

Crime incidents are strongly clustered around urban centres and areas of high economic activity, particularly in London, West Midlands, and West Yorkshire, while suburban regions such as Surrey and Thames Valley show more dispersed patterns.

From a real estate perspective, this indicates that crime risk is highly localised rather than uniform across regions. Even within higher-density urban areas, there are pockets of lower crime that may remain attractive to buyers, while properties near concentrated hotspots may face reduced desirability.

Overall, spatial crime patterns highlight the importance of location-specific analysis when assessing property value, buyer appeal, and investment potential.

10. Key Insights

The analysis highlights several important insights relevant to property sales and location desirability.

Firstly, total crime volume is strongly influenced by population size and urban density. While London records the highest number of crimes, this does not necessarily indicate the highest risk for residents. This distinction is critical when evaluating property attractiveness, as buyers may respond more to perceived safety than absolute crime levels.

Secondly, when crime is standardised by population, regions such as West Midlands and West Yorkshire demonstrate higher crime rates. These areas may present greater challenges for property sales, as higher crime exposure can reduce buyer confidence, impact property values, and increase perceived risk.

Thirdly, the composition of crime is consistent across regions, with violence and theft-related offences dominating. These crime types are particularly important in real estate, as they are more likely to influence buyer decision-making, insurance costs, and long-term investment appeal.

Spatial analysis further shows that crime is concentrated in urban centres and commercial hubs. Properties located within or near these high-density clusters may face lower desirability, while areas with more dispersed crime patterns (e.g. suburban regions) may be more attractive to buyers seeking safer environments.

Overall, the findings suggest that crime rate, crime type, and spatial concentration are key factors influencing property desirability, and should be considered alongside traditional market indicators when making sales and investment decisions.

11. Recommendations

Based on the exploratory data analysis, it is recommended that the Metropolitan Police Service and West Midlands Police areas are prioritised for further investigation, due to their distinct crime patterns and relevance to property sales strategy.

The Metropolitan Police Service represents the UK's largest and most dynamic property market, with the highest overall crime volume and strong spatial clustering in central urban areas. While crime levels are influenced by population density, the concentration of incidents highlights significant variation in desirability within London itself. A more granular analysis (e.g. borough or neighbourhood level) would enable identification of high-demand, lower-risk areas, supporting more targeted pricing, marketing, and investment strategies.

The West Midlands Police area is recommended due to its elevated relative crime exposure and clear clustering around key urban centres such as Birmingham. Rather than indicating uniformly high risk, this pattern suggests strong variation between locations within the region, where some areas may experience higher levels of crime while others remain more stable and attractive.

This creates an opportunity to:

- Identify lower-risk neighbourhoods within a major urban region
- Target areas with strong growth potential and more accessible property prices
- Support strategic positioning of properties in emerging or regenerating locations

From a real estate perspective, these two regions provide a valuable comparison between:

- A high-demand, high-density global city (London)
- A large, multi-centre urban region with varied local market conditions (West Midlands)

Further investigation into these areas will support:

- Identification of desirable vs less desirable locations at a granular level
- More informed pricing, marketing, and sales strategies
- A clearer understanding of how crime patterns influence buyer behaviour and investment decisions

12. Limitations

This analysis is subject to several limitations that should be considered when interpreting the findings for real estate decision-making:

- **Reported crime data only**: The dataset includes only reported crimes, meaning some incidents may go unreported. This may lead to an underestimation of actual crime levels and affect the accuracy of regional comparisons.

- **Population-based crime rates**: Crime rates are calculated using residential population data and do not account for daytime population, tourism, or commuting patterns. This is particularly relevant for areas such as central London, where the working population exceeds the residential population, potentially overstating crime risk.

- **Lack of socio-economic context**: The dataset does not include factors such as income levels, housing conditions, or regeneration activity. These are important in real estate, as areas with higher crime may still be attractive due to economic growth or investment potential.

- **Approximate spatial data**: Crime locations are based on approximate coordinates rather than exact locations, which may slightly reduce the accuracy of identifying precise crime hotspots for micro-level property decisions.

- **Limited time period**: The analysis covers two years of data, which may not capture longer-term trends or structural changes in crime that could influence long-term property investment decisions.